

Hasti KabirSamadi

📍 Tehran, Iran 🌐 hastikabirsamadi.github.io ✉ hastikabirsamadi@aut.ac.ir ☎ +98 935 411 6985
🔗 github.com/Hastikabirsamadi 🏫 Amirkabir University

Profile

I am a Computer Engineering senior at Amirkabir University of Technology (Tehran Polytechnic) with strong interest in bioinformatics and NLP. I have hands-on experience processing and analyzing RNA sequencing data and sentiment analysis tasks, particularly fine-tuning transformer based language models. I have strong background in deep learning and I am proficient in python libraries (Pytorch, Pandas, SciPy, NumPy, etc). I am passionate about applying machine learning to biological and medical challenges and looking for opportunities in biotechnology where I can grow my expertise and contribute to impactful projects.

Education

B.Sc. in Computer Engineering 2022 – present
Amirkabir University of Technology (Tehran Polytechnic) | Iran
GPA (last three semesters): 18.5 / 20
Farzanegan 2 High School 2017 – 2022
National Organization for Development of Exceptional Talents of Iran

Honors & Awards

Ranked 71st in the nationwide Iranian Universities Entrance Examination 2022
among more than 145,000 participants

Research Experience

Research Assistant, Amirkabir University of Technology 2026 – present
Supervisor: Prof. Fatemeh Zare-Mirakabad (Dept. of Mathematics & Computer Science)

- Evaluating RNA language models (ERNIE-RNA, RiNALMo, and related architectures) across structural, functional, and evolutionary tasks in a zero-shot setting.
- Investigating structural and functional signals encoded in transformer attention matrices and contextual embedding representations of RNA sequences.
- Developing a novel probing method that extracts task-relevant predictions solely from attention matrices and embedding matrices — without fine-tuning or downstream task-specific heads.

Research Intern, Iran Telecommunication Research Center (ITRC) Summer 2025

- Investigated multilingual transformer models for Persian sentiment classification.
- Fine-tuned XLM-RoBERTa on Persian Twitter datasets and evaluated performance.
- Conducted error analysis, model comparison, and documented results in a reproducible pipeline.

Academic Projects

scArches-Plus: Transfer-Learning Framework with Correlation Penalty for Single-Cell Atlas Mapping 2025
Bioinformatics course project, Amirkabir University of Technology

- Re-implemented scArches with a Correlation Penalty regulariser → clustering purity +8%.
- Added ExtEncoder layer & redesigned Decoder → batch-effect correction -22% (Silhouette score +0.11).
- Enabled transfer-learning fine-tuning without raw reference data → training time -40%.

Research Interests

- **Bioinformatics & Computational Biology:**
Single-cell and bulk RNA-seq analysis, gene regulatory network inference, multi-omics data integration, and ML-driven approaches to genomics and healthcare data.
- **Deep Learning for Biological Sequences:**
Applying transformer and foundation model architectures to DNA, RNA, and protein sequence analysis; probing attention and embedding representations; transfer learning for biological atlas mapping.
- **Natural Language Processing:**
Fine-tuning and evaluating transformer-based models for sequence classification, covering the full pipeline from dataset preparation and tokenization to supervised fine-tuning and evaluation.

Related Courses and Workshops

Advanced Bioinformatics Sharif University of Technology (online)	in progress
RNA-seq Data Analysis Workshop Genome Guide (online)	2026
Fundamentals of Bioinformatics Amirkabir University of Technology	2024
Introduction to NLP: Basic Concepts, Preprocessing and Representations Amirkabir University of Technology	2023

Skills & Languages

- **Programming:** Python, C, C++, Java.
- **ML & Data:** PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib.
- **Bioinformatics:** Scanpy, AnnData, scArches; scRNA-seq data processing; RNA-seq analysis pipelines.
- **Tools & Platforms:** Git & GitHub, Docker, Linux/Bash, $\LaTeX$.
- **Languages:** Persian (Native), English (Advanced).

Teaching Experience

Head Teaching Assistant, Microprocessors & Assembly Language (Undergraduate) • Amirkabir University of Technology • Lead and coordinated the TA team: assigned tasks, distributed responsibilities, ensured consistent grading standards. • Planned assessments (assignments, quizzes, projects) and oversaw QA of questions, model answers, and rubrics. • Served as primary intermediary between students and the instructor.	Fall 2025
Teaching Assistant — Amirkabir University of Technology • Computer Networks (Spring 2026) · Software Test (Spring 2026) · Software Engineering 2 (Fall 2025) • Microprocessors & Assembly Language (Spring 2024) · Advanced Programming (Fall 2024, Fall 2025) • Differential Equations (Spring 2023)	